Optical Information Processing

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What is optical information processing?

- Use of optical techniques to process data
- Purpose built hardware accelerators
- Think current GPUs (Graphics Processing Units), but optical based
- Very fast
- Very little energy

Current GPUs

The best we can do?

- Electronic devices using logic gates
- Connect to computers through standard ports (PCIe)
- Perform matrix maths extremely quickly
- Mostly used for graphics rendering
- Recently repurposed for AI calculations

New ideas

Optical digital logic gates

- Take advantage of optical nonlinearities in ring resonators
- Use them to modulate optical signals
- Cascadable, extremely low energy logic gates have been demonstrated

In the future

“Coherent Ising Machine”

- A type of neural network used to solve optimisation problems
- Matrix-Vector products are calculated efficiently using programmable Mach-Zender Interferometer meshes
- These calculations are used to simulate the neural network

Right now
Lightmatter

- Lightmatter sell an Optical AI Accelerator
- Uses standard interconnects
- Performs AI calculations 5 times faster than Nvidia
- 7 times more energy efficient

References

- https://www.photonicssociety.org/images/files/publications/Newsletter/Photonics_Aug2021-web.pdf
- https://lightmatter.co/

Thank you!

- Questions?